

Erdős Problem #254

EP-254

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subseteq \mathbb{N}$ be such that $|A \cap [1, 2x]| - |A \cap [1, x]| \rightarrow \infty$ as $x \rightarrow \infty$ and $\sum_{n \in A} \{ \theta n \} = \infty$ for every $\theta \in (0, 1)$, where $\{x\}$ is the distance of x from the nearest integer. Then every sufficiently large integer is the sum of distinct elements of A .

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>